

Dependency-Colored Concentration for Finite-Alphabet Shared-Exclusions PID

Finite-sample law control, drift separation, local atom moduli,
and a formal evidence boundary

pid-rs project analysis

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Abstract

This document gives project-defined validation, new in `pid-rs`, for the categorical shared-exclusions partial information decomposition (SxPID) defined by Makkeh, Gutknecht, and Wibral. The paper-defined functional is not new. The document derives a finite-sample concentration bound for the complete empirical joint law when all complete rows share one common finite law and a fixed coloring partitions observations into classes whose complete rows are mutually independent. Dependence across classes can be arbitrary. A class-size proxy that is optimal within the declared Hölder–Hoeffding proof scheme improves the ordinary color-count factor for unbalanced classes. A telescoping error allocation gives a time-uniform envelope. A nonidentical-law extension separates random error from explicit drift. A stronger common-support modulus then transfers joint-law error to pointwise and averaged SxPID atoms. The local result includes one- Λ cumulative control, general-source Möbius-row bounds, and complete two-source atom bounds. Exact diamond analysis reduces the synergy modulus to $\Lambda - \eta$. Lean checks deterministic algebraic subclaims, and an executable challenge suite retains counterexamples to weaker assumptions. The result does not validate continuous shared exclusions, arbitrary time series, adaptive evaluation transforms, or binary64 asymptotics.

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1 Status, scope, and provenance

The mathematical functional, the concentration ingredients, and this repository artifact have different origins.

Object	Origin	Status in this document
Categorical SxPID	Paper-defined by Makkeh, Gutknecht, and Wibral; its lattice has a formal-logic basis due to Gutknecht, Wibral, and Makkeh [1, 2]	Implemented for two to four sources; receives a finite-sample bound and an almost-sure exact-real consistency implication when the displayed envelope vanishes
Hoeffding bounded-variable inequality	Paper-defined classical result [3]	Used inside each independent color class
Dependency-color concentration	Published background includes Janson’s coloring method and later Hölder formulations [4, 5]	Project-defined proof composition with a class-size proxy that is optimal within the declared Hölder–Hoeffding proof scheme; no novelty claim
Empirical-law L^1 deviation	Published independent and identically distributed (i.i.d.) background includes Weissman et al. [6]	One-sided subset union bound under the stated color contract
SxPID local modulus	Project-defined validation of the paper-defined functional	One- Λ cumulative control, general-source Möbius-row bounds, and sharper complete two-source atom-specific bounds. The exact ordinary-diamond diameter and a sharp conditioned-diamond bound reduce the synergy modulus to $\Lambda - \eta$. Endpoint constructions and one bounded law fixture give only local evidence for other coefficients; no result proves global sharpness of the complete atom or averaged bounds
Drift envelope	Project-defined extension	Concentration about the average row law plus explicit bias to a reference law

The machine-readable authority is `method-catalog.json`. The rendered authority is `METHODS.md`. The Rust implementation routes covered by the final composition are the stable categorical functions

`discrete_sxpid2`, `discrete_sxpid2_averaged`, `discrete_sxpid3`,
`discrete_sxpid3_averaged`, `discrete_sxpid_n`, and `discrete_sxpid_n_averaged`.

They are exported from `stable::categorical`. This document adds no estimator and no public Rust API. Its proof and executable artifacts are validation evidence.

2 Finite setting and the dependence contract

Let $Z_1, \dots, Z_n$ take values in a finite alphabet $\mathcal{Z}$ with $|\mathcal{Z}| = K$. For SxPID, one row is the complete joint symbol

$$Z_i = (S_{1i}, \dots, S_{mi}, T_i). \quad (1)$$

Dependence between source and target coordinates inside a row is unrestricted. It is the signal that PID analyzes.

For the common-law theorem, assume every row has marginal law P . Let a deterministic color map partition the indices into occupied classes $C_1, \dots, C_r$. Define

$$n_j = |C_j| > 0, \quad n = \sum_{j=1}^r n_j. \quad (2)$$

The complete rows $\{Z_i : i \in C_j\}$ must be mutually independent for every j . Dependence across different classes can be arbitrary. The color map must not depend on the observed row values.

Define

$$\hat{P}_n(z) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{Z_i = z\}, \quad V_n = \left(\sum_{j=1}^r \sqrt{n_j} \right)^2, \quad d_{\text{eff},n} = \frac{V_n}{n}. \quad (3)$$

Proposition 2.1 (Effective-color range). *The deterministic proxy obeys*

$$n \leq V_n \leq rn, \quad 1 \leq d_{\text{eff},n} \leq r. \quad (4)$$

Proof. For the lower bound, expand the square of a sum of nonnegative square roots and retain its diagonal terms. For the upper bound, apply Cauchy–Schwarz:

$$\left(\sum_j \sqrt{n_j} \right)^2 \leq \left(\sum_j 1^2 \right) \left(\sum_j n_j \right) = rn. \quad (5)$$

The Lean artifact proves both inequalities over exact real numbers. $\square$

The value $d_{\text{eff},n}$ is a proof constant derived from the declared class sizes. It is not an estimated effective sample size.

3 Finite-sample law concentration

Theorem 3.1 (Dependency-colored empirical-law bound). *Assume $K \geq 2$ and the contract in Section 2. Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left(\left\|\hat{P}_n - P\right\|_1 \geq \varepsilon\right) \leq \min\left\{1, (2^K - 2) \exp\left(-\frac{n^2 \varepsilon^2}{2V_n}\right)\right\}. \quad (6)$$

If at most d colors are occupied, then

$$\mathbb{P}\left(\left\|\hat{P}_n - P\right\|_1 \geq \varepsilon\right) \leq \min\left\{1, (2^K - 2) \exp\left(-\frac{n \varepsilon^2}{2d}\right)\right\}. \quad (7)$$

Proof. Fix a nonempty proper subset $A \subset \mathcal{Z}$. Put

$$Y_i = \mathbf{1}\{Z_i \in A\} - P(A), \quad S_j = \sum_{i \in C_j} Y_i. \quad (8)$$

Let $R = \sum_j \sqrt{n_j}$ and select the Hölder exponents

$$p_j = \frac{R}{\sqrt{n_j}}, \quad \sum_j \frac{1}{p_j} = 1. \quad (9)$$

Each $p_j \geq 1$ because $R \geq \sqrt{n_j}$, so these are valid Hölder exponents. For $\lambda > 0$, generalized Hölder gives

$$\mathbb{E} \exp \left(\lambda \sum_j S_j \right) \leq \prod_j \left(\mathbb{E} e^{p_j \lambda S_j} \right)^{1/p_j}. \quad (10)$$

Rows are mutually independent inside each color. Hoeffding's lemma for a centered random variable with range length one gives

$$\mathbb{E} e^{p_j \lambda S_j} \leq \exp \left(\frac{n_j p_j^2 \lambda^2}{8} \right). \quad (11)$$

Therefore,

$$\mathbb{E} \exp \left(\lambda \sum_j S_j \right) \leq \exp \left(\frac{\lambda^2 V_n}{8} \right). \quad (12)$$

Chernoff's method, optimized at $\lambda = 4nt/V_n$, yields

$$\mathbb{P} \left(\hat{P}_n(A) - P(A) \geq t \right) \leq \exp \left(-\frac{2n^2 t^2}{V_n} \right). \quad (13)$$

For any signed finite measure with total mass zero,

$$\frac{1}{2} \left\| \hat{P}_n - P \right\|_1 = \max_{A \subseteq \mathbb{Z}} \{ \hat{P}_n(A) - P(A) \}. \quad (14)$$

Set $t = \varepsilon/2$ and take a union bound over the $2^K - 2$ nonempty proper subsets. This proves Equation (6). Complement subsets already encode the opposite sign, so no additional factor of two is required. Proposition 2.1 gives Equation (7). $\square$

For $K = 1$, the L^1 distance is identically zero, and the finite-sample formulas reduce to the true but empty statement $0 \leq 0$ for every $\varepsilon > 0$. The time-uniform radius contains $\log(2^K - 2)$, so Section 4 requires $K \geq 2$. The subset factor is exponential in K , so the bound can be vacuous on a large alphabet. This result makes no sharpness or sample-complexity claim. If $s = |\text{supp}(P)| \geq 2$ is known before the data are inspected, then $2^s - 2$ can replace $2^K - 2$. An observed support size is not a known population support size. When $s = 1$, the law distance is zero and the result is vacuous.

Proposition 3.2 (Hölder proxy optimum). *For $q_j > 0$ with $\sum_j q_j^{-1} = 1$,*

$$V_n \leq \sum_j n_j q_j. \quad (15)$$

Equality holds for $q_j = R/\sqrt{n_j}$.

Proof. The exponent conditions force $q_j \geq 1$ for every j . Thus, they are valid Hölder exponents. Cauchy–Schwarz gives

$$\left(\sum_j \sqrt{n_j} \right)^2 = \left(\sum_j \sqrt{n_j q_j} \sqrt{q_j^{-1}} \right)^2 \leq \left(\sum_j n_j q_j \right) \left(\sum_j q_j^{-1} \right). \quad (16)$$

The selected exponents make the two Cauchy–Schwarz vectors proportional. $\square$

This is an optimum inside the Hölder–Hoeffding proof scheme for this partition. It is not a claim that V_n is the best possible constant for every admissible joint law.

4 Time-uniform envelope and consistency

Assume $K \geq 2$. For an infinite sequence, require one fixed map from the positive integers to a finite or countable color set. Only finitely many colors can be occupied in each prefix. The entire infinite collection in each color must consist of mutually independent complete rows. Let V_n use the nonempty class sizes in the first n rows. A uniform bound of at most d occupied colors is required only for the coarse radius in Equation (23).

Fix $0 < \alpha < 1$ and define

$$C_K = 2^K - 2, \quad \alpha_n = \frac{\alpha}{n(n+1)}. \quad (17)$$

The allocation has the exact finite identity

$$\sum_{n=1}^N \alpha_n = \alpha \left(1 - \frac{1}{N+1}\right) \leq \alpha. \quad (18)$$

Lean proves the unit-scale finite identity by induction. Multiplication by the fixed scalar α gives Equation (18).

Define

$$R_n = \sqrt{\frac{2V_n}{n^2} \log\left(\frac{C_K n(n+1)}{\alpha}\right)}, \quad (19)$$

$$D_n = \min\{2, R_n\}. \quad (20)$$

Theorem 4.1 (Anytime law envelope). *Under the infinite-sequence contract,*

$$\mathbb{P}(\forall n \geq 1 : \|\hat{P}_n - P\|_1 \leq D_n) \geq 1 - \alpha. \quad (21)$$

Proof. For every n , substitution in Theorem 3.1 gives

$$C_K \exp\left(-\frac{n^2 R_n^2}{2V_n}\right) = \alpha_n. \quad (22)$$

The Lean artifact checks the algebraic exponent cancellation. Positivity, the square-root substitution, and its probability interpretation remain in this prose proof. A union bound and Equation (18) give simultaneous control by R_n . Every probability-law L^1 distance is also at most two. Thus it is at most $D_n = \min\{2, R_n\}$. $\square$

The coarser d -color radius is

$$D_n^{(d)} = \min\left\{2, \sqrt{\frac{2d}{n} \log\left(\frac{C_K n(n+1)}{\alpha}\right)}\right\}. \quad (23)$$

Under one uniform fixed bound d , this is $O(\sqrt{d \log(n)/n})$. For each positive rational error threshold, the tails in Equation (7) are summable. Borel–Cantelli and a countable intersection over those thresholds therefore give

$$\|\hat{P}_n - P\|_1 \longrightarrow 0 \quad \text{almost surely.} \quad (24)$$

If the number of colors grows, the displayed envelope converges to zero under the sufficient condition

$$\frac{V_n \log n}{n^2} \longrightarrow 0. \quad (25)$$

This condition is not necessary for empirical-law convergence. For example, i.i.d. rows still converge when a needlessly fine singleton coloring gives $V_n = n^2$. The condition is sufficient for this declared-color envelope. A singleton coloring also permits complete dependence across rows, so the theorem cannot infer convergence from that coloring alone.

Corollary 4.2 (Fixed-width overlapping windows). *Let E_t be i.i.d. innovations. Fix $w \in \mathbb{N}$ before evaluation, and let*

$$Z_t = f(E_t, \dots, E_{t+w-1}) \quad (26)$$

for one fixed measurable finite-output map f . Coloring row t by $t \bmod w$ satisfies the dependence contract with at most $d = w$ colors.

Proof. Two rows in the same color use disjoint innovation blocks. Every finite collection in that color is therefore mutually independent. The fixed map and the i.i.d. innovation law give the common row law. $\square$

This corollary covers a fixed categorical overlapping-window construction from i.i.d. innovations. It does not cover circular wraparound windows, a width selected from evaluation outcomes, a rolling-start sequence of repeated claims, or innovations with unspecified time dependence.

Definition 4.3 (Strong ℓ -dependence). Let $\ell \in \mathbb{N}_0$. An ordered sequence is strongly ℓ -dependent when

$$\sigma(Z_i : i \leq t) \text{ is independent of } \sigma(Z_i : i \geq t + \ell + 1) \quad (27)$$

for every t .

Coloring index t by $t \bmod (\ell + 1)$ separates consecutive members of each color by more than ℓ . Repeatedly separate the last row from the sigma-field generated by all preceding rows. This ordered induction proves mutual independence of all complete rows in each color. Pairwise lag independence alone does not prove this property. The common-law theorem still requires one common row law. If row laws differ, the drift theorem and its additional premises apply.

5 Nonidentical row laws and explicit drift

Assume $K \geq 2$. Let row i have law P_i , and keep the within-color mutual-independence contract. Define

$$\bar{P}_n = \frac{1}{n} \sum_{i=1}^n P_i. \quad (28)$$

Center the subset indicator for row i by $P_i(A)$. The Hölder–Hoeffding proof is unchanged.

Theorem 5.1 (Average-law and reference-law bounds). *For every $\varepsilon > 0$,*

$$\mathbb{P}\left(\left\|\hat{P}_n - \bar{P}_n\right\|_1 \geq \varepsilon\right) \leq \min\left\{1, C_K \exp\left(-\frac{n^2 \varepsilon^2}{2V_n}\right)\right\}. \quad (29)$$

For an explicit reference law $P_\star$, put

$$b_n = \left\|\bar{P}_n - P_\star\right\|_1. \quad (30)$$

Then

$$\mathbb{P}\left(\left\|\hat{P}_n - P_\star\right\|_1 \geq b_n + \varepsilon\right) \leq \min\left\{1, C_K \exp\left(-\frac{n^2 \varepsilon^2}{2V_n}\right)\right\}. \quad (31)$$

Proof. Equation (29) follows from the row-specific centering. The triangle inequality gives

$$\left\|\hat{P}_n - P_\star\right\|_1 \leq \left\|\hat{P}_n - \bar{P}_n\right\|_1 + b_n, \quad (32)$$

which proves Equation (31). $\square$

If $\|P_i - P_\star\|_1 \leq \beta_i$, then convexity gives

$$b_n \leq \frac{1}{n} \sum_{i=1}^n \beta_i. \quad (33)$$

Under one fixed infinite row-law sequence and coloring with the same premises, the all-prefix reference-law envelope is

$$\mathbb{P}(\forall n \geq 1 : \|\hat{P}_n - P_\star\|_1 \leq \min\{2, b_n + D_n\}) \geq 1 - \alpha. \quad (34)$$

The bias term is necessary. Concentration about $\bar{P}_n$ does not identify a fixed scientific estimand.

6 Strengthened common-support local SxPID continuity

This section records claim SXPID-TV-001, revision 2. It gives project-defined validation of a paper-defined functional. It does not define a new PID measure or estimator. All statements use exact real arithmetic.

Let p and q be laws on one fixed complete joint alphabet. Define

$$\delta = \|q - p\|_1, \quad \eta = \frac{\delta}{2}, \quad p_{\min} = \min_{z \in \text{supp}(p)} p(z), \quad L = \log(1/p_{\min}). \quad (35)$$

Assume

$$\text{supp}(q) \subseteq \text{supp}(p), \quad 0 \leq \delta < 2p_{\min}. \quad (36)$$

The convention $\eta = \sup_E |q(E) - p(E)|$ is total variation. Equation (36) implies equal support. Put

$$p_s = (1 - s)p + sq, \quad \mu_s = p_{\min} - s\eta, \quad 0 \leq s \leq 1. \quad (37)$$

Every supported cell under p_s has mass at least $\mu_s > 0$.

Lemma 6.1 (Path oscillation). *Let $\Delta = q - p$. For every real-valued function g on the finite common support,*

$$\left| \sum_z \Delta(z) g(z) \right| \leq \eta \left(\max_z g(z) - \min_z g(z) \right). \quad (38)$$

If a differentiable functional F satisfies $\text{osc } \nabla F(p_s) \leq 1/\mu_s$, then

$$|F(q) - F(p)| \leq \int_0^1 \frac{\eta}{p_{\min} - s\eta} ds = \log \frac{p_{\min}}{p_{\min} - \eta} =: \Lambda(\delta, p_{\min}). \quad (39)$$

Proof. Subtract the midpoint of the range of g in Equation (38) and use $\sum_z \Delta(z) = 0$. Apply that inequality to the directional derivative of F along p_s . The strict support margin makes the complete segment positive, so the fundamental theorem of calculus and direct integration give Equation (39). $\square$

Fix a supported realization $z = (s_1, \dots, s_m, t)$ and a nonempty antichain α of nonempty source subsets. Define

$$A_\alpha = \bigcup_{a \in \alpha} \{S_a = s_a\}, \quad B_\alpha = A_\alpha \cap \{T = t\}, \quad C = \{T = t\}. \quad (40)$$

By definition, $B_\alpha = A_\alpha \cap C$, and the keyed realization belongs to B_α . For a law Q , define

$$c_\alpha^+(z; Q) = -\log Q(A_\alpha), \quad (41)$$

$$c_\alpha^-(z; Q) = \log \frac{Q(C)}{Q(B_\alpha)}, \quad (42)$$

$$c_\alpha^{\text{sx}}(z; Q) = \log \frac{Q(B_\alpha)}{Q(A_\alpha)Q(C)}. \quad (43)$$

Proposition 6.2 (One- Λ cumulative modulus). *For $u \in \{+, -, \text{sx}\}$,*

$$|c_\alpha^u(z; q) - c_\alpha^u(z; p)| \leq \Lambda. \quad (44)$$

Proof. Write $A = A_\alpha$ and $B = B_\alpha$. The cell-gradient values are

Region	∇c^+	∇c^-	∇c^{sx}
B	$-1/Q(A)$	$1/Q(C) - 1/Q(B)$	$1/Q(B) - 1/Q(A) - 1/Q(C)$
$A \setminus C$	$-1/Q(A)$	0	$-1/Q(A)$
$C \setminus A$	0	$1/Q(C)$	$-1/Q(C)$
outside $A \cup C$	0	0	0

The three gradient diameters are at most $1/Q(B)$. For the net column, compare every pair and use $Q(B) \leq Q(A), Q(C)$. Since $p_s(B) \geq \mu_s$, Lemma 6.1 proves the result. The direct net calculation avoids a nonoptimal factor of two. $\square$

Let M be the fixed Möbius matrix of the complete redundancy lattice. Define

$$\pi_\alpha^u(z; Q) = \sum_\beta M_{\alpha\beta} c_\beta^u(z; Q), \quad r_\alpha = \sum_\beta |M_{\alpha\beta}|, \quad (45)$$

$$\bar{\pi}_\alpha^u(Q) = \sum_{z \in \text{supp}(Q)} Q(z) \pi_\alpha^u(z; Q). \quad (46)$$

Also define

$$h = \log \frac{2}{1 + p_{\min}}, \quad J = L - 2h \geq 0. \quad (47)$$

Theorem 6.3 (General-source atom bounds). *For every complete-lattice node and $u \in \{+, -, \text{sx}\}$,*

$$|\pi_\alpha^u(z; q) - \pi_\alpha^u(z; p)| \leq r_\alpha \Lambda. \quad (48)$$

The averaged component bounds are

$$|\bar{\pi}_\alpha^\pm(q) - \bar{\pi}_\alpha^\pm(p)| \leq \min \{L, r_\alpha \Lambda + \eta L\}. \quad (49)$$

The averaged net bound is

$$|\bar{\pi}_\alpha^{\text{sx}}(q) - \bar{\pi}_\alpha^{\text{sx}}(p)| \leq \min \{2L, r_\alpha \Lambda + \delta \min \{L, r_\alpha (L - h)\}\}. \quad (50)$$

Proof. Equation (48) follows from Proposition 6.2 and the triangle inequality. Makkeh, Gutknecht, and Wibral prove that every informative and misinformative pointwise atom is nonnegative on the complete finite antichain lattice [1, Theorem IV.3 and Appendix A]. Their base-2 sign result is unchanged after conversion to nats. Each component atom is therefore at most its top-node component sum, which is at most L . Hence, under p ,

$$0 \leq \pi_\alpha^+, \pi_\alpha^- \leq L, \quad -L \leq \pi_\alpha^{\text{sx}} \leq L. \quad (51)$$

For a cumulative net value, write $a = p(A_\alpha)$, $b = p(B_\alpha)$, and $c = p(C)$. Then $a, c \geq b \geq p_{\min}$ and $a + c - b \leq 1$. Thus

$$ac \leq \frac{(a+c)^2}{4} \leq \frac{(1+b)^2}{4}, \quad (52)$$

and

$$-J \leq \log \frac{b}{ac} \leq L. \quad (53)$$

The cumulative range width is $2(L - h)$. A Möbius row gives net-atom oscillation at most $2r_\alpha(L - h)$; Equation (51) also gives oscillation at most $2L$.

Split an average change into a pointwise-value term and a probability-weight term. Center the second term as in Equation (38). The component weight term is at most ηL . The net weight term is at most $\delta \min\{L, r_\alpha(L - h)\}$. This proves the inner bounds. For the outer caps, the common support has at most $1/p_{\min}$ cells. Each averaged component atom is nonnegative and at most its top component sum, which is a source entropy or conditional source entropy. It is therefore at most L under either law. Each averaged net atom lies in $[-L, L]$. $\square$

The lattice's sole bottom Möbius row has $r_\alpha = 1$, so its net weight coefficient is $L - h$. Every nonbottom row is a nonzero integer vector with zero sum. Thus it has at least one positive and one negative entry, so $r_\alpha \geq 2$. Since $J = L - 2h \geq 0$, its inner net coefficient simplifies to L .

Theorem 6.4 (Complete two-source improvement). *Define*

$$\Lambda_{\text{syn}} = \log \frac{p_{\min}}{p_{\min} - \eta} - \eta = \Lambda - \eta. \quad (54)$$

For redundancy and either unique atom, and for $u \in \{+, -, \text{sx}\}$,

$$|\pi_\alpha^u(z; q) - \pi_\alpha^u(z; p)| \leq \Lambda. \quad (55)$$

For synergy and the same three component labels,

$$|\pi_{\text{syn}}^u(z; q) - \pi_{\text{syn}}^u(z; p)| \leq \Lambda_{\text{syn}}. \quad (56)$$

Put

$$J_q = \log \frac{(1 + p_{\min} - \eta)^2}{4(p_{\min} - \eta)}. \quad (57)$$

The averaged bounds are

<i>Atom</i>	<i>Component average change</i>	<i>Net average change</i>
<i>Redundancy</i>	$\min\{L, \Lambda + \eta L\}$	$\min\{2L, \Lambda + \delta(L - h)\}$
<i>Either unique atom</i>	$\min\{L, \Lambda + \eta L\}$	$\min\{2L, \Lambda + \delta(L - h)\}$
<i>Synergy</i>	$\min\{L, J_q, \Lambda_{\text{syn}} + \eta J\}$	$\min\{2L, J + J_q, \Lambda_{\text{syn}} + \delta J\}$

Proof. Let $E_i = \{S_i = s_i\}$, $U = E_1 \cup E_2$, and $C = \{T = t\}$. Redundancy uses one event log, one nested log ratio, or one intersection-PMI expression, so Proposition 6.2 applies.

For one unique atom, put

$$N(a, b) = \log \frac{a + b}{a}. \quad (58)$$

Its two nonzero gradient values are

$$D_a = \frac{1}{a + b} - \frac{1}{a}, \quad D_b = \frac{1}{a + b}, \quad (59)$$

whose diameter is $1/a$. For a path law $Q = p_s$, put

$$x_a = Q(E_i \cap C), \quad x_b = Q((U \setminus E_i) \cap C), \quad (60)$$

$$y_a = Q(E_i \cap C^c), \quad y_b = Q((U \setminus E_i) \cap C^c). \quad (61)$$

The keyed cell gives $x_a \geq \mu_s > 0$. Addition is componentwise. Define

$$G_N(x, y) = N(x + y) - N(x). \quad (62)$$

For $i \in \{a, b\}$, the y_i -gradient coordinate is $F_i = D_i(x + y)$ and the x_i -gradient coordinate is $X_i = F_i - D_i(x)$. A coordinate outside the two event regions has gradient value zero.

Put

$$A = \frac{1}{x_a}, \quad B = \frac{1}{x_a + x_b}, \quad E = \frac{1}{x_a + y_a}, \quad P = \frac{1}{x_a + x_b + y_a + y_b}. \quad (63)$$

The five values are

$$F_a = P - E, \quad F_b = P, \quad X_a = P - E - B + A, \quad X_b = P - B, \quad 0. \quad (64)$$

Their minimum is $\min\{F_a, X_b\}$, their maximum is $\max\{F_b, X_a\}$, and their exact diameter is

$$\max\{E, B, A - B, A - E\}. \quad (65)$$

The full-union reciprocal P cancels. Each displayed term lies in $[0, A]$, so the diameter is at most $1/x_a$. Equivalently, the F_a – F_b difference is at most $1/(x_a + y_a) \leq 1/x_a$, and the same-index differences $F_i - X_i = D_i(x)$ have magnitude at most $1/x_a$. Also,

$$X_a - X_b = \frac{1}{x_a} - \frac{1}{x_a + y_a}, \quad (66)$$

and the two crossed differences are

$$F_a - X_b = -\frac{1}{x_a + y_a} + D_b(x), \quad F_b - X_a = \frac{1}{x_a + y_a} + D_a(x). \quad (67)$$

Each crossed difference is a difference of two values in $[0, 1/x_a]$. Each of F_a, F_b, X_a, X_b has magnitude at most $1/x_a$, which also controls every pair with the outside zero coordinate. Thus all five gradient values have diameter at most $1/x_a$.

Lean proves the candidate extrema, Equation (65), pairwise domination, and attainment. It also proves that no positive quantity can be subtracted uniformly from $1/x_a$ over the nonnegative algebraic mass domain. Its boundary witness sets $x_b = y_a = y_b = 0$ and compares the full-event exclusive coordinate with the outside coordinate. This zero-side-mass construction is an algebraic gradient witness. It is not by itself a supported common-law perturbation or an SxPID-realizability result. The realizable two-cell atom counterexamples below separately show that unique information can attain the unrefined Λ modulus.

For synergy, partition the source space into $E_1 \cap E_2$, $E_1 \setminus E_2$, $E_2 \setminus E_1$, and the outside region $O = (E_1 \cup E_2)^c$. For any law, let a, b, c, o be the masses of these four regions in that order. Define

$$\Phi(a, b, c) = \log \frac{(a+b)(a+c)}{a(a+b+c)}. \quad (68)$$

Its gradient values, including the outside region o , are

$$D_a = \frac{1}{a+b} + \frac{1}{a+c} - \frac{1}{a} - \frac{1}{a+b+c}, \quad (69)$$

$$D_b = \frac{1}{a+b} - \frac{1}{a+b+c}, \quad D_c = \frac{1}{a+c} - \frac{1}{a+b+c}, \quad D_o = 0. \quad (70)$$

Here $D_a \leq 0 \leq D_b, D_c$ and

$$D_b - D_a = \frac{1}{a} - \frac{1}{a+c}, \quad D_c - D_a = \frac{1}{a} - \frac{1}{a+b}. \quad (71)$$

If $b \leq c$, then D_b is the maximum and D_a is the minimum. If $c \leq b$, then D_c is the maximum and D_a is the minimum. Therefore,

$$\text{osc } D = \frac{1}{a} - \frac{1}{a + \max\{b, c\}}. \quad (72)$$

For a probability partition, $a + \max\{b, c\} \leq a + b + c \leq 1$, so

$$\text{osc } D \leq \frac{1}{a} - 1. \quad (73)$$

The informative synergy component is a diamond function on the full source-region masses. The misinformative component is a diamond function on nonnegative target-restricted region masses whose total is at most one.

For a path law $Q = p_s$, define

$$(x_a, x_b, x_c, x_o) = (Q(E_1 \cap E_2 \cap C), Q((E_1 \setminus E_2) \cap C), Q((E_2 \setminus E_1) \cap C), Q(O \cap C)), \quad (74)$$

and define (y_a, y_b, y_c, y_o) by replacing C with C^c . The keyed cell gives $x_a \geq \mu_s > 0$. Addition is componentwise, and

$$G_\Phi(x, y) = \Phi(x + y) - \Phi(x). \quad (75)$$

For $i \in \{a, b, c, o\}$, the y_i -gradient coordinate is $F_i = D_i(x + y)$ and the x_i -gradient coordinate is $X_i = F_i - D_i(x)$. These are the eight gradient values.

The ordinary-diamond signs give

$$F_a \leq 0 \leq F_b, F_c, \quad X_a \geq F_a, \quad X_b \leq F_b, \quad X_c \leq F_c. \quad (76)$$

Both outside coordinates are zero. Hence

$$\text{diam}(F, X) = \max\{F_b, F_c, X_a\} - \min\{F_a, X_b, X_c\}. \quad (77)$$

Put

$$A = \frac{1}{x_a}, \quad B = \frac{1}{x_a + x_b}, \quad C = \frac{1}{x_a + x_c}, \quad D = \frac{1}{x_a + x_b + x_c}, \quad (78)$$

$$E = \frac{1}{x_a + y_a}, \quad P = \frac{1}{x_a + x_b + y_a + y_b}, \quad Q = \frac{1}{x_a + x_c + y_a + y_c}, \quad R = \frac{1}{S}, \quad (79)$$

where $S = x_a + x_b + x_c + y_a + y_b + y_c$. The nine candidate maximum-minus-minimum differences are

$$F_b - F_a = E - Q, \quad F_c - F_a = E - P, \quad X_a - F_a = A + D - B - C, \quad (80)$$

$$F_b - X_b = B - D, \quad F_c - X_c = C - D, \quad X_a - X_b = A - C + Q - E, \quad (81)$$

$$X_a - X_c = A - B + P - E, \quad F_b - X_c = P - Q + C - D, \quad F_c - X_b = Q - P + B - D. \quad (82)$$

The reciprocal orders are

$$R \leq P, Q \leq E \leq A, \quad P \leq B, \quad Q \leq C, \quad D \leq B, C \leq A, \quad (83)$$

and reciprocal supermodularity on the base diamond gives $B + C \leq A + D$. These relations bound the first seven differences by $A - R$. For $F_b - X_c$, if $P \leq Q$, use $P - Q + C - D \leq C - D \leq A - R$. If $Q < P$, use $P \leq B$, $B + C \leq A + D$, and $R \leq Q$. The other crossed difference is symmetric. Therefore,

$$\text{diam}(F, X) \leq \frac{1}{x_a} - \frac{1}{S}. \quad (84)$$

All nine extremal regimes in Equation (77) can occur, even when all six displayed masses are positive. There is no one fixed-pair formula. The universal bound in Equation (84) is attained.

Every probability assignment has $S \leq 1$. Since the keyed cell gives $x_a \geq \mu_s$, Equations (73) and (84) bound each synergy gradient diameter by $1/\mu_s - 1$. Centered total variation and path integration give

$$\eta \int_0^1 \left(\frac{1}{p_{\min} - s\eta} - 1 \right) ds = \Lambda_{\text{syn}}. \quad (85)$$

This proves Equation (56). The redundancy and unique calculations above prove Equation (55). The synergy modulus is nonnegative because $-\log(1 - \eta/p_{\min}) \geq \eta/p_{\min} \geq \eta$.

The synergy improvement cannot be applied to the other atoms. Start with two cells of mass $1/2$ and move mass η from the first to the second. Redundancy informative and net use cells $(0, 0, 0)$ and $(1, 1, 1)$; redundancy misinformative uses $(0, 0, 0)$ and $(1, 1, 0)$. Source-1 unique informative and net use $(1, 1, 1)$ and $(0, 1, 0)$; its misinformative component uses $(1, 1, 1)$ and $(0, 1, 1)$. Exchange the sources for source-2 unique information. In each case, the stated first-cell atom is $-\log p(z)$, so its change is exactly $\log[(1/2)/(1/2 - \eta)] = \Lambda > \Lambda_{\text{syn}}$.

Formal boundary, counterexample, and sharpness. Lean proves the exact ordinary-diamond diameter and its attainment. It also proves that the candidate extrema in Equation (77) are attained and give the exact conditioned-diamond diameter. The theorem `conditioned_diamond_lifted_gradient_refined_coordinate_diameter_le` checks Equation (84) over the exact real numbers. It assumes only

$$x_a > 0, \quad x_b, x_c, y_a, y_b, y_c \geq 0. \quad (86)$$

It needs no positive exclusive-region mass, outside-region mass, normalization, or upper bound on the displayed masses. The theorem `conditioned_diamond_lifted_gradient_refined_bound_attained` proves equality for $x = (x_a, 0, 0)$ and $y = (0, 0, S - x_a)$ when $0 < x_a \leq S$. Neither theorem formalizes differentiation, path integration, probability events, or the identification with net SxPID synergy. The following declarations check the $1/x_a - 1$ pairwise and absolute-coordinate corollaries when the six displayed masses total at most one:

```
conditioned_diamond_lifted_gradient_probability_domain_coordinate_diameter_le
conditioned_diamond_lifted_gradient_probability_domain_absolute_coordinate_le
```

Separate validity of the base and full diamonds is insufficient if a negative exclusive complement-region lift mass is allowed. Let

$$x = (1/4, 1/4, 0), \quad y = (0, -1/4, 1/2). \quad (87)$$

The base and full vectors x and $x + y = (1/4, 0, 1/2)$ are separately valid diamonds, but $y_b < 0$. The lifted coordinates include $F_b = 8/3$ and $X_c = -2$. Their difference is $14/3$, which exceeds both the endpoint-only refined claim $8/3$ and the coarse claim $1/x_a = 4$. The exact fixture retains this counterexample.

The fixture also retains the source-exchanged negative-exclusive case. A third case uses

$$x = (1/2, 0, 0), \quad y = (-1/4, 3/4, 0). \quad (88)$$

The base and full diamonds are valid, but the coordinate diameter is $3 > 2 = 1/x_a$ and also exceeds the endpoint-only refined claim 1. Thus separate endpoint validity does not replace the componentwise nonnegative lift premise.

The refined coefficient is attained. For

$$x = (\varepsilon, 0, 0), \quad y = (0, 0, 1 - \varepsilon), \quad 0 < \varepsilon < 1, \quad (89)$$

the displayed mass total is one, and $F_b - X_c = (1 - \varepsilon)/\varepsilon$ equals $1/x_a - 1/S$. Its ratio to the refined bound is one. Its ratio to the older $1/x_a$ bound is $1 - \varepsilon$. The fixture uses $\varepsilon = 1/1000$ and

the source-swapped case. The Python generator and the Rust test independently reconstruct all eight coordinates, the refined bound, and all 64 ordered differences in each of seven rational cases. Nine additional assignments have six positive displayed masses that sum to one and realize all nine extremal regimes. The checks also reconstruct the exact ordinary-diamond and conditioned-nested identities on the same inputs. One case has zero lift. One algebra-only case has displayed mass total 41 and checks that normalization is not used by the algebraic theorem. It is not a probability assignment.

It remains to control the weight terms. For net unique information, let a, b be the target-region masses and let $a_f \geq a$, $b_f \geq b$ be the full-event masses. Then

$$\exp(\pi_{\text{unique}}^{\text{sx}}) = \frac{a(a_f + b_f)}{a_f(a + b)} \geq \frac{4a}{(1 + a)^2} \geq \frac{4p_{\min}}{(1 + p_{\min})^2}. \quad (90)$$

The first inequality uses $b_f \geq b$, $a_f + b_f \leq 1$, and $(1 - b)(a + b) \leq (1 + a)^2/4$. Under the p -law, this gives the $-J$ lower endpoint for each unique net atom. The upper endpoint is L because $N(a_f, b_f) \leq L$ and $N(a, b) \geq 0$. Moreover,

$$0 \leq \Phi(a, b, c) \leq J \quad \text{under the } p\text{-law.} \quad (91)$$

For the upper bound, put $u = a + b + c \geq a$. The arithmetic-geometric mean inequality gives $(a + b)(a + c) \leq (a + u)^2/4$. The ratio $(a + u)^2/(4au)$ is nondecreasing in $u \geq a$, so $u \leq 1$ reduces it to $(1 + a)^2/(4a)$. This last expression is nonincreasing on $(0, 1]$. Under the p -law, redundancy and unique net atoms lie in $[-J, L]$, and net synergy lies in $[-J, J]$. Every q -law cell has mass at least $p_{\min} - \eta$, so the same proof gives $0 \leq \Phi \leq J_q$ under q . These ranges control the p -weighted term in the average-change split and give the endpoint caps J_q and $J + J_q$. The other outer caps use support cardinality and therefore bound the averaged values under both laws. Centered total variation now gives the displayed atom-specific weight terms. This two-source result uses only the displayed log expressions and elementary range arguments. It does not use the published general-source component-nonnegativity result used in Theorem 6.3. $\square$

The lower endpoint in Equation (53) is attained as an event bound when $p_{\min} \leq 1/3$, the intersection has mass $p_{\min}$, and the two exclusive regions each have mass $(1 - p_{\min})/2$. A separate event construction attains the upper endpoint when $p_{\min} \leq 1/2$. These endpoint laws do not prove that a general Möbius-row bound is sharp.

The revision-1 constants $(\Lambda, 2\Lambda, 3\Lambda)$ for cumulative terms remain valid triangle-inequality baselines. They are superseded because they ignore the direct nested and intersection-PMI gradients. A generic two-point range extremizer retained in the executable record is not shown to be SxPID-realizable and proves no SxPID sharpness claim.

For a realizable two-source family, in lexicographic binary-cell order let

$$p_\varepsilon = \left(\varepsilon, \frac{1}{6}, \frac{1}{3}, \varepsilon, \frac{1}{2} - 5\varepsilon, \varepsilon, \varepsilon, \varepsilon \right), \quad 0 < \varepsilon < \frac{1}{12}. \quad (92)$$

Net synergy at 000 and 111 equals $-L + \log 10 + o(1)$ and $L - \log(40/3) + o(1)$. Their oscillation divided by L tends to two. Moving mass $0 < \rho < \varepsilon$ from 000 to 111 proves asymptotic necessity of the δL centered net-weight coefficient for that decomposition. For admissible $\rho < \varepsilon$, the Λ_{syn} pointwise-value term can dominate the total bound. Thus this family does not prove global sharpness of the complete average bound.

The asymptotic statement is a prose derivation from the displayed family. The executable fixture checks only the finite case $\varepsilon = 1/60$. Its largest synergy misinformative and net changes are approximately $0.9806\Lambda_{\text{syn}}$ and $0.9597\Lambda_{\text{syn}}$.

Under the standing conditions $0 \leq \delta$, $0 < p_{\min} \leq 1$, and $\delta \leq p_{\min}/2$, put $\xi = \delta/(2p_{\min}) \leq 1/4$. The Lean theorem `refined_log_modulus_linearized_chain` proves

$$0 \leq \Lambda_{\text{syn}} \leq \Lambda \leq \frac{\xi}{1-\xi} \leq \frac{4\xi}{3} = \frac{2\delta}{3p_{\min}}. \quad (93)$$

For numerical evaluation, require finite inputs with $0 < p_{\min} \leq 1$ and $0 \leq \eta < p_{\min}$. First compute $q_{\text{floor}} = p_{\min} - \eta$ in the working precision. Reject the input if this value is not finite and strictly positive. One formula is not stable over the complete strict-support range. When $p_{\min}$ and q_{floor} are subnormal and $q_{\text{floor}} \ll p_{\min}$, the rounded ratio $\eta/p_{\min}$ can lose most of the remaining-floor information.

If $r = \eta/p_{\min} \leq 1/2$, evaluate $\Lambda = -\log(1-r)$ as `-log1p(-r)` or as $r + \sum_{k=2}^{\infty} r^k/k$. The committed test-only evaluator uses the positive-series form. Do not subtract nearby binary64 values. Use

$$\Lambda_{\text{syn}} = [-\log(1-r) - r] + r(1-p_{\min}). \quad (94)$$

Evaluate the bracket with a compensated `log1pmx` routine or the positive series $\sum_{k=2}^{\infty} r^k/k$. The second term avoids both reciprocal overflow and the cancellation in $\eta(1/p_{\min} - 1)$.

If $r > 1/2$, use

$$u = \frac{q_{\text{floor}}}{p_{\min}}, \quad \Lambda = -\log u, \quad \Lambda_{\text{syn}} = \Lambda - \eta. \quad (95)$$

Use $q_{\text{floor}}/p_{\min}$ directly. Under this bounded input contract, $p_{\min}/q_{\text{floor}} \leq 2^{53}$, so the inverse quotient cannot overflow. The direct quotient keeps the documented normal interval and avoids an unnecessary inversion. Do not subtract two large logarithms when $p_{\min}$ is very small, because that route can lose relative accuracy. For binary64 inputs on this upper branch, $p_{\min}/2 < \eta < p_{\min}$. Sterbenz subtraction therefore makes $q_{\text{floor}} = p_{\min} - \eta$ exact. The smallest positive represented gap gives $u \geq 2^{-53}$, so u is normal. IEEE binary64 round-to-nearest, ties-to-even division keeps the computed quotient below $1/2$ for the contracted inputs. The bounded evaluator enforces $2^{-53} \leq u \leq \text{nextDown}(1/2)$ and does not accept $u = 1/2$. To see the strict result, upper-branch selection implies the exact ratio $\eta/p_{\min} > 1/2 + 2^{-54}$; a midpoint tie would round to $1/2$. Exact subtraction then gives $q_{\text{floor}}/p_{\min} < 1/2 - 2^{-54}$ before the second division is rounded. This argument assumes round-to-nearest, ties-to-even operations, no excess-precision double rounding, and gradual underflow at the subnormal boundary. The committed bounded test checks the quotient precondition and uses runtime canaries for the rounding mode and gradual underflow before it uses the quotient-log route. The generator converts the upper-branch inputs to exact binary fractions. It rejects a represented floor that is not exactly $p_{\min} - \eta$. This binary64 argument is numerical guidance, not part of the exact-real theorem. The branch at $r = 1/2$ is an implementation choice.

Evaluate the endpoint ceiling without first forming its quotient. If $q_{\text{floor}} \leq 1/2$, use

$$J_q = 2 \log(1 + q_{\text{floor}}) - \log q_{\text{floor}} - \log 4. \quad (96)$$

If $q_{\text{floor}} > 1/2$, put $t = (1 - q_{\text{floor}})/(1 + q_{\text{floor}})$ and use

$$J_q = -\log(1 - t^2) \quad (97)$$

with a `log1p` operation. The first route avoids overflow for a subnormal floor. The second avoids cancellation as the floor tends to one.

The committed generator uses 400-digit Decimal arithmetic. It evaluates Λ with one high-precision quotient logarithm. The Rust reconstruction is a bounded fixture check for the committed laws and numerical stress cases, not a general interval implementation. No result in this section is a binary64 error theorem, a support-creation theorem, or a continuous-estimator theorem.

7 Composition with SxPID

For common-law sampling, empirical support is a subset of population support almost surely. On the event in Theorem 4.1, the strict condition

$$D_n < 2p_{\min} \quad (98)$$

forces empirical and population support to agree. The realized error satisfies $\delta \leq D_n$. Both pointwise moduli are nondecreasing in δ . In particular,

$$\frac{\partial \Lambda_{\text{syn}}}{\partial \delta} = \frac{1}{2(p_{\min} - \delta/2)} - \frac{1}{2} \geq 0. \quad (99)$$

The quantity J_q is also nondecreasing because its floor argument decreases with δ . Thus every displayed right-hand side is nondecreasing. Substitute D_n for δ everywhere, including in η , Λ_{syn} , and J_q . Use the result in Equations (44), (48), (49), (50), and (55), (56), together with the two-source table. This gives simultaneous exact-real envelopes for every supported cumulative term, every fixed general-source Möbius atom, and every complete two-source atom.

A usable numerical envelope needs a justified lower bound on $p_{\min}$. The observed minimum frequency is not such a bound. A positive population cell can be absent from a finite sample.

For Theorem 5.1, also require every P_i to be supported inside $\text{supp}(P_*)$. Define

$$p_{\min, \star} = \min_{z \in \text{supp}(P_*)} P_{\star}(z). \quad (100)$$

If

$$b_n + D_n < 2p_{\min, \star}, \quad (101)$$

then the realized error is at most $b_n + D_n$. Monotonicity gives the same composition with $b_n + D_n$ on the right-hand side.

A frozen finite-output transform can be handled conditionally on an independent training artifact. Conditional on that artifact, the color map must be fixed. Every row must have the stated conditional law. The complete rows in every color class must be mutually independent. A random conditional support floor gives a conditional result unless a deterministic lower bound is available.

The displayed common-law envelope proves almost-sure exact-real finite-alphabet SxPID plug-in consistency under the sufficient condition $V_n \log(n)/n^2 \rightarrow 0$; fixed d is sufficient. The displayed drift envelope proves almost-sure exact-real reference-law SxPID consistency when this sufficient condition holds and $b_n \rightarrow 0$. These are not necessary conditions under a stronger sampling theorem. The result does not validate continuous $I_{\cap}^{\text{sx}}$, arbitrary overlapping windows, adaptive preprocessing on evaluation rows, unknown mixing rates, or binary64 asymptotics.

8 Counterexamples and invalidated routes

Invalid route	Counterexample	Required correction
Pairwise independence inside one color	Let U be uniform on $\mathbb{F}_2^4$. Use the 15 nonzero linear forms $a \cdot U$ as binary rows. Every pair is independent. All rows are zero with probability $1/16$. The empirical L^1 error is then one, while the false one-color bound is $2e^{-15/2} \approx 0.00111$.	Require mutual independence of all complete rows in each color.
Remove the color factor	Take m independent fair bits and copy each bit once into each of d colors. Each color is i.i.d., but only $m = n/d$ independent values remain.	Keep V_n , or keep the coarser factor d .
Infer convergence from one color per row	Set every row equal to one common fair bit. Singleton colors satisfy their local premise, but $V_n = n^2$ and the empirical law does not converge to the fair law.	This coloring cannot prove convergence; use a stronger coloring or another theorem.
Read the envelope condition as necessary	Let the rows be i.i.d. but assign each row a separate color. Then $V_n = n^2$, although the empirical law converges by the strong law.	State $V_n \log(n)/n^2 \rightarrow 0$ only as a sufficient condition for this envelope.
Data-adaptive coloring without the complete conditional contract	Set every row equal to one fair bit, then select one occupied color from that bit. Conditional rows are degenerate, but their conditional law is not the common unconditional law.	Require fixed colors, or prove the full conditional law and independence premises.
Substitute a mixing label for the color premise	Use a stationary fair two-state Markov chain with flip probability $\theta = 1/100$. Its transition dependence decays geometrically. For two rows, the empirical L^1 error is one with probability $1 - \theta = 0.99$, above the false one-color bound $2e^{-1}$.	Use a theorem with an explicit mixing coefficient and rate, or prove a valid independence coloring.
Pairwise lag independence as strong ℓ -dependence	The finite-field construction has pairwise independence and a joint failure.	Require the sigma-field definition. Residues modulo $\ell + 1$ then give valid colors by ordered induction.
Halve the net weight term from the generic range alone	On two points, let $w = (-\delta/2, \delta/2)$ and a generic row value be $f = (-r_\alpha L, r_\alpha L)$. Then $ \sum wf = r_\alpha L\delta$. This row value is not shown to be realizable by an SxPID atom when $r_\alpha > 1$. It proves no SxPID sharpness claim.	Keep the generic range baseline unless a separate SxPID argument supplies a smaller range.

Invalid route	Counterexample	Required correction
Replace Λ by Λ_{syn} for every two-source atom	The displayed two-cell SxPID laws make a redundancy or unique component change by exactly $\Lambda > \Lambda_{\text{syn}}$ when $\eta > 0$.	Use Λ_{syn} only for synergy. Keep Λ for redundancy and unique information.
Allow $\delta = 2p_{\min}$	Move all mass from a least-probable supported cell. Its required logs become undefined.	Keep $\delta < 2p_{\min}$ strict.
Control only separate marginals	Set $S_1 = S_2 = X$. For $0 < \rho < 1$, at the supported realization $(X, T) = (0, 0)$, use fair binary X, T marginals with diagonal cells $(1 + \rho)/4$ and off-diagonal cells $(1 - \rho)/4$, then replace ρ by $-\rho$. Marginals stay fixed, but pointwise redundancy changes by $\log((1 + \rho)/(1 - \rho))$.	Control the complete joint law.
Allow new support with a linear average bound	Let $p(0, 0, 0) = 1$, and let $q(0, 0, 0) = 1 - \varepsilon$, $q(1, 1, 1) = \varepsilon$. Then $\ q - p\ _1 = 2\varepsilon$, while the averaged redundancy is $-(1 - \varepsilon) \log(1 - \varepsilon) - \varepsilon \log \varepsilon$.	Require common support or use a nonlinear boundary modulus.
Use observed minimum frequency as a population floor	An unobserved positive cell has empirical frequency zero.	Use an external floor or a separate simultaneous lower-confidence result.

The executable challenge suite enumerates the finite-field pairwise-independence construction, copied-color construction, singleton-color construction, and adaptive-color construction. It also checks a support-deletion boundary, an unspecified-mixing construction, a generic net-weight range extremizer, a univariate-marginal-only construction, and a new-support construction. Three cases show that endpoint validity does not replace the componentwise nonnegative lift premise. Seven rational cases check the eight conditioned-diamond coordinates and all 64 ordered differences per case. They include a zero-lift boundary and one explicitly algebra-only, unnormalized case. Nine cases have six positive displayed masses that sum to one and realize all nine exact minimum/maximum regimes. The same exact inputs also check the ordinary-diamond identity and the conditioned-nested candidate and closed-form diameter identities. Six two-cell SxPID cases attain Λ for non-synergy components and violate the false all-atom refinement. The remaining checks cover the telescoping allocation, class-size inequalities, high-precision one-log cases, and all displayed bounds on six committed two-source law pairs. Two pairs make the J_q component cap and the $J + J_q$ net cap active. Ten binary64 cases challenge the adaptive refined-modulus evaluation. Stored hexadecimal payloads bind parsed operands and subtraction results. The cases include adjacent inputs around $r = 1/2$, the exact seam, the exact lower endpoint $q_{\text{floor}}/p_{\min} = 2^{-53}$ of the upper route, an extreme normal-scale input, and a subnormal near-boundary floor. Six cases challenge the endpoint-ceiling evaluation. They include adjacent inputs around $q_{\text{floor}} = 1/2$, the exact seam, a floor near one, the exact $q_{\text{floor}} = 1$ positive-zero endpoint, and the smallest positive subnormal floor. Their 400-digit references use the exact real values represented by the parsed binary64 inputs.

The generator stores the $q_{\text{floor}} = 1$ endpoint as the exact algebraic value zero. It does not derive that endpoint by subtracting rounded logarithms. The bounded evaluator also returns this endpoint directly. It does not depend on the signed-zero result of a logarithm call. One full-support pair has count vectors

$$(10, 100, 200, 10, 250, 10, 10, 10) \quad \text{and} \quad (1, 100, 200, 10, 250, 10, 10, 19). \quad (102)$$

It has $\delta = 3/100$, $\eta = 3/200$, $p_{\min} = 1/60$, and $p_{\min}/(p_{\min} - \eta) = 10$. Its largest observed synergy misinformative and net changes exceed $0.98\Lambda_{\text{syn}}$ and $0.959\Lambda_{\text{syn}}$. This is bounded near-tightness evidence for the synergy pointwise constant, not a proof of global sharpness.

The fixed-window fixture uses independent fair innovations U_t, V_t and

$$S_{1,t} = U_t \text{ OR } U_{t+1}, \quad S_{2,t} = V_t \text{ OR } V_{t+1}, \quad T_t = S_{1,t} \text{ XOR } S_{2,t}. \quad (103)$$

Even and odd starts are the two colors. Its exact joint count weights are $1 : 3 : 3 : 9$. The high-precision Decimal oracle supplies expected logarithmic values. For stored logarithmic constants and bounds, the Rust reconstruction tolerance is $32\varepsilon_{\text{mach}} \max(1, |x_{\text{Rust}}|, |x_{\text{oracle}}|)$. For categorical estimator outputs, the absolute comparison ceiling is $32\varepsilon_{\text{mach}}$ nats, where $\varepsilon_{\text{mach}} = 2^{-52}$ for binary64. The test also checks that row reversal gives bit-identical output.

9 Formal and executable evidence boundary

The Lean project is pinned to one toolchain and one exact dependency manifest. The repository checker rejects admitted proof placeholders, builds the project, and replays its declarations with Lean’s kernel checker.

The checked Lean project proves the following results across the `Dependence.lean` and `LocalContinuity.lean` source files:

- the event-mass $\delta/2$ lemma and exact attainment by the positive-coordinate event;
- positivity under the strict support margin;
- the one-log modulus, its monotonicity, and its rational simplification;
- the centered zero-sum oscillation inequality;
- reciprocal-diameter bounds for negative-event-log, nested-log-ratio, and intersection-PMI gradient coordinates;
- the exact ordinary two-source diamond gradient diameter, its attainment, its union-reciprocal corollary, and the algebraic and logarithmic $0 \leq \Phi \leq J$ bounds;
- the five-coordinate conditioned nested-event candidate extrema, exact closed-form gradient diameter, attainment, zero-side-mass algebraic witness, and impossibility of a positive uniform subtraction on that algebraic domain;
- the exact candidate-extrema form and the union-reciprocal upper bound for the eight-coordinate conditioned-diamond gradient, plus its normalized probability-domain corollaries, under one positive common mass and nonnegative exclusive and complement-region masses; an exact-real equality witness and exact fixtures establish sharpness;
- the complete small-error linearization chain for Λ_{syn} under $0 \leq \delta \leq p_{\min}/2$ and $0 < p_{\min} \leq 1$;
- the effective-color proxy bounds and the normalized factor range;
- the generic absolute linear-row bound;
- component and net range transfer, centered weight bounds, and combined finite-average perturbation bounds;
- the factorized positive segment-event floor;
- the exact unit-scale finite telescoping allocation; and
- the algebraic radius-exponent cancellation.

Lean does not encode random variables or conditional laws. It does not encode generalized Hölder, Hoeffding’s lemma, the Chernoff argument, or the probability union bound. The drift

result and fixed-window independence argument also remain outside Lean. Differentiation, path integration, and the identification of event masses with algebraic gradient coordinates remain outside Lean. Lean does not encode SxPID events, the redundancy lattice, the published pointwise component-nonnegativity theorem, or the identification of its generic lemmas with SxPID atoms. Rust refinement and binary64 rounding remain outside Lean. The probability theorem is a prose proof. Lean machine-checks only the deterministic obligations listed above. The Python suite uses exact rational arithmetic for finite probability and count identities, plus 400-digit Decimal arithmetic for logarithms and an absolute 10^{-350} tolerance for its internal Decimal identity checks. It checks all 64 ordered conditioned-diamond coordinate pairs in each of seven rational cases, all nine exact extremal regimes, and three endpoint-valid negative-lift counterexamples. Transcendental logarithms are not exact rational values. From each local count-table pair, the Rust test independently reconstructs both empirical laws, δ , η , $p_{\min}$, Λ , Λ_{syn} , L , h , J , J_q , the atom family, and every stored bound before it checks the implementation outputs. A separate bounded test reconstructs the ten adaptive-modulus and six endpoint-ceiling numerical cases and requires selected naive routes to fail. For this stability test only, an expected zero must match positive zero bit for bit. Each other comparison uses

$$256\varepsilon_{\text{mach}} \max(|x_{\text{oracle}}|, s_{\text{op}}), \quad (104)$$

where s_{op} is the bounded sum of the absolute displayed operation terms for that case. A selected naive route must be nonfinite or differ from the oracle by more than 1024 times that tolerance. The separate law reconstruction and categorical-output tolerances remain the $32\varepsilon_{\text{mach}}$ rules declared above. These checks are bounded executable evidence. They are not a general proof, a global binary64 bound, or external review.

10 Ecosystem use boundary

Consumer	Result supplied here	Still required
Prisoma	Categorical SxPID law control after a frozen finite map when held-out rows meet the color contract	Concrete row law, justified support floor, independent transform receipt, and a valid window coloring
Galadriel	Dependence-aware categorical alternative to an i.i.d. plug-in claim	Removal of same-window fitting and generic jitter routes; explicit observation and coloring contracts
Haldir	Mathematical target for block or residue-class designs	Direct dependency, corrected atom interpretation, and proof of strong ℓ -dependence or another valid coloring
Crebain	Finite-alphabet validation component for a future integration	Exact observation mapping, run-log contract, direct dependency, and explicit scientific estimand

This document does not assert that a consumer meets these premises. Integration remains `not_claimed` until the consumer records and checks them.

11 Reproduction

From the repository root, run

```
python3 scripts/check-lean-finite-convergence.py
```

```
python3 scripts/generate-dependency-colored-sxpid-oracle.py
cargo test --locked -p pid-core --test dependency_colored_sxpid_oracle
scripts/check-dependency-colored-sxpid-pdf.sh
```

The PDF checker rebuilds this source with a fixed source-date epoch, rejects LaTeX diagnostics, and compares the exact PDF bytes. The Lean checker binds the toolchain, manifest, source set, dependency revisions, dependency origins, and clean checkout state. It removes ambient Git routing and global or system configuration while retaining each dependency checkout's local origin configuration for verification.

12 Conclusion

The result supplies a dependence-aware finite-sample path from a declared finite-row coloring to exact-real categorical shared-exclusions PID bounds. Its displayed envelope proves almost-sure exact-real consistency under the sufficient condition $V_n \log(n)/n^2 \rightarrow 0$. The corresponding reference-law result also uses $b_n \rightarrow 0$. These are not necessary conditions under a stronger sampling theorem. The result separates four objects that must remain separate: sampling fluctuation, deterministic drift, support margin, and program arithmetic. The color-size proxy improves the proof constant for unbalanced classes. The telescoping allocation gives a simple time-uniform statement. The local logarithmic modulus gives a direct law-to-atom transfer once common support is justified. Every weaker route listed in Section 8 is retained as invalidated work.

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